

Start-Up Transient Simulation of a Liquid Rocket Engine

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During the start-up of a liquid rocket engine most of the engine components are required to work at extreme operating conditions; this phase features very complex phenomena such as combustion instabilities, water hammer effects in feed lines, turbopumps operating far from design conditions and two-phase flows that cannot be neglected when design studies of an engine are to be performed or the assessment of the valve opening sequence has to be done. For these reasons simulations of the ignition transient phase become necessary in order to reduce the experimental tests and increase the engine safety and reliability.

The subject of the present work is to develop an LRE transient model, to apply it to simulate the start-up transient and to validate the model with an existing liquid rocket engine (the RL-10).

A system approach has been used, since it is necessary to take into account all the interactions between all the components of an engine. This choice is fundamental if a detailed estimation of the engine transient behaviour is the task. The system simulated by the code encompasses the complete feeding system from the tanks down to the turbopump subsystems, the valves, the pipes, the thrust chamber's inlets, the flow ducts of the thrust chamber itself and the chemical reactions inside the combustion chamber.

The engine model is built using ESPSS,⁵ the propulsion system library compatible with EcosimPro, an object oriented tool capable of modelling various kinds of dynamic systems. The engine model includes also original models developed by the authors and presented in previous works⁶ such as a new injector plate model, and more realistic heat transfer coefficient correlations.

The transient phase of the Pratt&Whitney RL-10 engine has been taken as validation case. This engine has been chosen for several reasons, among which the good availability of engine construction data, performance and tests results in open literature.¹⁻³

The main subsystem models of this engine will be described and their validation will be presented. Then the integrated model of the entire engine will be described alongside the comparison of the simulation with experimental data.

Nomenclature

A	Area, m ²
C_o	Turbine exit sound velocity, m/s
C_p	Specific heat, J/kg·K
H	Flow total enthalpy, kJ/kg
h	Dimensionless pump head, -
h_c	Convective heat transfer coefficient, W/m ² ·K
M	Propellants mixture ratio, -
n	Rotational speed ratio, -
N	Turbine velocity ratio, -
$\dot{m}$	Mass flow rate, kg/s
Q	Volumetric mass flow rate, m ³ /s
Q^+	Turbine mass flow coefficient, -
$\dot{q}$	Heat flux, W/m ²
r	Turbine blade mean radius, m
ST	Turbine specific torque, -
St	Stanton number, -

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T	Temperature, K
TDH	Pump total dynamic head, m
β	Dimensionless pump torque, -
θ	Dimensionless pump variable, -
λ	Thermal conductivity, W/m·K
ν	Volumetric flow rate ratio, -
ξ	Friction coefficient, -
Π	Turbine total to total pressure ratio, -
ρ	Density, kg/m ³
σ	Radiative heat transfer coefficient, W/m ² ·K ⁴
τ	Mechanical torque, N·m
ω	Rotational speed, rad/s
<i>Subscript</i>	
aw	adiabatic wall
R	rated condition
rad	radiative
ref	reference condition
th	throat condition
w	wall

I. Introduction

This paper deals with the development and the applications of a computational transient model of the RL-10A-3-3A rocket engine. The RL-10 engine is based on an expander cycle, in which the fuel is used to cool the main combustion chamber and the thermal energy added to the fuel drives the turbopumps. The RL-10 rocket engine is an important component of the American space infrastructure. Two RL-10 engines form the main propulsion system for the Centaur upper stage vehicle, which boosts commercial, scientific and military payloads from a high altitude into Earth orbit. The RL-10A-3-3A developed by Pratt & Whitney under contract to NASA, incorporates component improvements with respect to the initial RL-10A-1 engine.

A cryogenic expander cycle engine involves a strong coupling between the different subsystems. This coupling is even stronger during the start and shut-down transients, when non-linear interactions between subsystems play a major role. In addition complex phenomena such as combustion, heat transfer, turbopump operation, phase change, valve maneuverings are concerned, as well as important changes in the thermodynamic properties of the fluids involved. The development of a computational transient model is therefore necessary to analyse and assess correctly the engine operation. More generally a transient model helps to reduce the number of engine tests by allowing to perform a certain amount of parametric studies in advance of the test campaign, and thus plays an important role in the cost and risk reduction.

The RL-10 engine has been used extensively as object of simulations in the past years.^{1-3,9,10} The work presented in this paper wants to show the improvements made in terms of modelling with respect to the other tools; indeed, the model present here features a 1-D discretisation not only in the cooling jacket model, but also for most of the other components, such as the combustion chamber, the venturi duct and the other pipes.

In previous works,³ the combustion chamber has been modelled as a built-in set of hydrogen/oxygen combustion tables. Here, a fully 1-D discretised chamber and nozzle features a chemical equilibrium model based on Gibbs energy minimization for each section along the chamber. The present model also contains an injector plate model representative not only of the capacitive effect of the injector dome mass but also of the convective and radiative heat fluxes from the chamber to the injector and of the conductive heat flux between the fuel and oxidiser injector domes. The thermal model used for the cooling jacket component is modelled as a “real” one and a half counterflow cooling jacket.

Finally it is important to mention that, to the best of the author’s knowledge, chill down and pre-start procedures were never simulated before with transient system tools. In the present work, the cool-down (pre-start) procedure has been simulated in order to obtain a accurate and complete engine state at start signal ($t=0$). The pre-start simulation results are in very good agreement with the few experimental data available.

II. Overview of the RL-10A-3-3A rocket engine

A. Engine architecture and operation

The RL10A-3-3A engine is an expander cycle engine, with nominal thrust of 73.4 kN, a mixture ratio around 5, a chamber pressure of 32.7 bar and a specific impulse of 444 seconds. The fuel turbine drives both the fuel and the oxidiser pumps (the latter being driven via a gear train). The presence of a bypass valve allows the adjustment of the flow rate through the turbine and is used consequently for thrust throttling; another valve present in the oxidiser line is used for mixture ratio trimming. The RL-10 engine has been designed to offer a multiple firing capability in flight. A schematic diagram of the engine is shown in Figure 1. The RL10A-3-3A includes seven engine valves as shown in Figure 2. The propellant flows to the engine can be shut off using the Fuel Inlet Valve (FINV) and the Oxidizer Inlet Valve (OINV). The fuel flow into the combustion chamber can be stopped by the Fuel Shut-off Valve (FSOV) located just upstream of the injector plenum. The FSOV is a helium operated, two position, normally closed, bullet-type annular gate valve. The valve serves to prevent fuel flow into the combustion chamber during the cool-down period and provide a rapid cut-off of fuel flow during engine shut-down.¹²

The fuel interstage and discharge cool-down valves (FCV-1 and FCV-2) are pressure-operated, normally open sleeve valves. The purposes of these valves are the following:¹²

- allow overboard venting of the coolant for fuel pump cool-down during engine pre-chill and pre-start
- provide first stage fuel pump bleed control during the engine start transient (for the FCV-1)
- provide fuel system pressure relief during engine shut-down

The Thrust Control Valve (TCV) is used to control thrust overshoot at start and maintain constant chamber pressure during steady-state operation. TCV is a normally closed, servo-operated, closed-loop, variable position bypass valve used to control engine thrust by regulation of turbine power. As combustion chamber pressure deviates from the desired value, action of the control allows the turbine bypass valve to vary the fuel flow through the turbine.¹²

The Oxidizer Control Valve (OCV) has two orifices; one regulates the main oxidizer flow (OCV-1) and the other controls the bleed flow required during engine start (OCV-2). The main-flow orifice in the OCV is actuated by the differential pressure across the LOX pump. The OCV valve is a normally closed, variable position valve. The valve controls oxidiser pump cool-down flow during the engine pre-start cycle and during engine start transient.¹²

The Venturi upstream of the turbine is designed to help stabilize the thrust control.

Ducts and manifolds in the RL10 are generally made out of stainless-steel and are not insulated.

B. Description of the start-up sequences

The RL-10 engine starts by using the pressure difference between the fuel tank and the nozzle exit (upper atmospheric pressure), and the ambient heat stored in the metal of the cooling jacket walls. The engine “bootstraps” to full-thrust within two seconds after ignition.

Before engine start, inlet valves are opened and propellants are allowed through the fuel pump for five seconds (cooled to prevent cavitation at engine start) and through the LOX system for nine seconds. This “pre-start” flow consumes approximately 10 kg of oxygen and 2.7 kg hydrogen.¹⁵ The fuel cool-down valves (see Figure 2) are open and the main shut-off valve (FSOV) is closed. The fuel flow is vented overboard through the cool-down valves and does not flow through the rest of the system; the latent heat in the metal of the combustion chamber cooling jacket is therefore available to help drive the start transient. The oxidiser pump is pre-chilled by a flow of oxygen, which passes through the Oxidiser Control Valve (OCV) and is vented through the combustion chamber and nozzle.

A typical plot of the valve movement during engine start is shown in Figure 3. To initiate start, the FSOV is opened and the fuel-pump discharge cool-down valve (FCV-2) is closed. The interstage cool-down valve (FCV-1) remains partially open in order to avoid stalling of the fuel pump during engine acceleration. The pressure drop between the fuel inlet and the combustion chamber drives fuel through the cooling jacket, picking up heat from the warm metal. This pressure difference also drives the heated fluid through the turbine, starting rotation of the pumps, which drive more propellant into the system. At start, the OCV also closes partially, restricting the flow of oxygen into the combustion chamber. This is done to limit

Table 1. RL-10A-3-3A construction data³

Name	Value	Units
<i>Fuel Turbopump</i>		
1st stage impeller diameter	179.6	mm
1st stage exit blade height	5.8	mm
2nd stage impeller diameter	179.6	mm
2nd stage exit blade height	5.588	mm
<i>Oxidiser Turbopump</i>		
Impeller diameter	106.7	mm
Exit blade height	6.376	mm
<i>Turbine</i>		
Mean line diameter	149.86	mm
Mass moment of inertia	0.008767	kg·m ²
<i>Cooling jacket</i>		
Number of short tubes	180	-
Number of long tubes	180	-
Channel width at throat	2.286	mm
Channel height at throat	3.556	mm
Total coolant volume	0.0158	m ³
Typical hot wall thickness	0.3302	mm
HGS effective surface area	4.645	m ²
<i>Thrust chamber</i>		
Chamber diameter	0.1303	m
Throat diameter	0.0627	m
Nozzle area ratio	61	-
Chamber/nozzle length	1.476	m
Number of injectors	216	-
Injector Assembly weight	6.72	kg

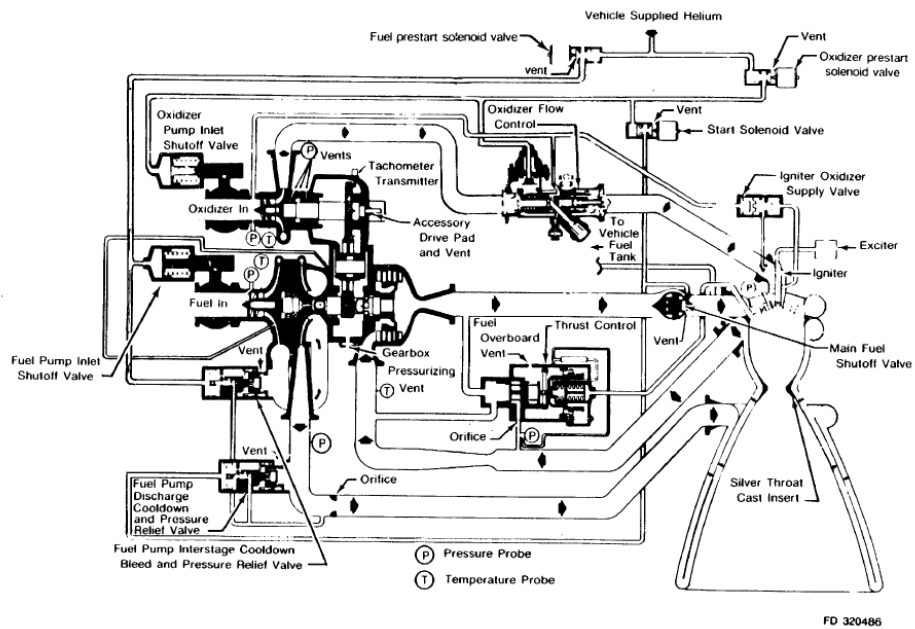


Figure 1. RL-10A-3-3A engine schematic¹⁵

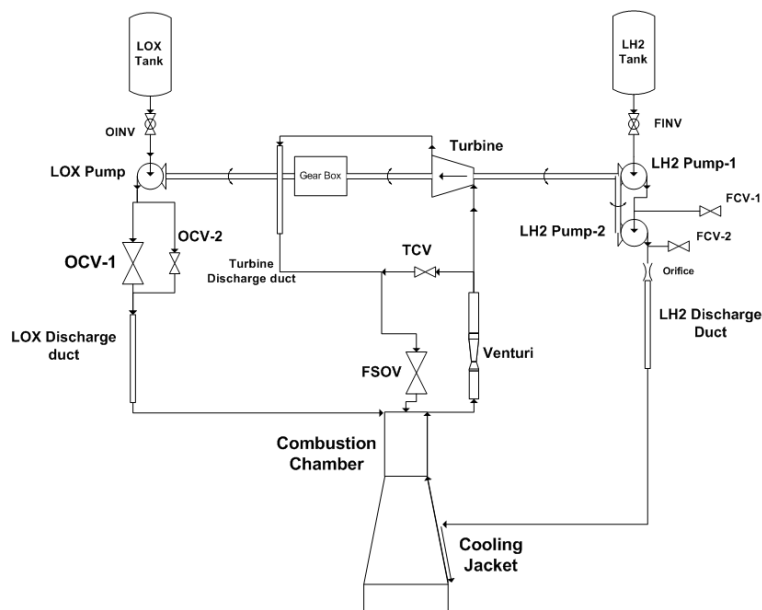


Figure 2. RL-10A-3-3A engine diagram

chamber pressure and ensure a forward pressure difference across the fuel turbine after ignition of the thrust chamber.

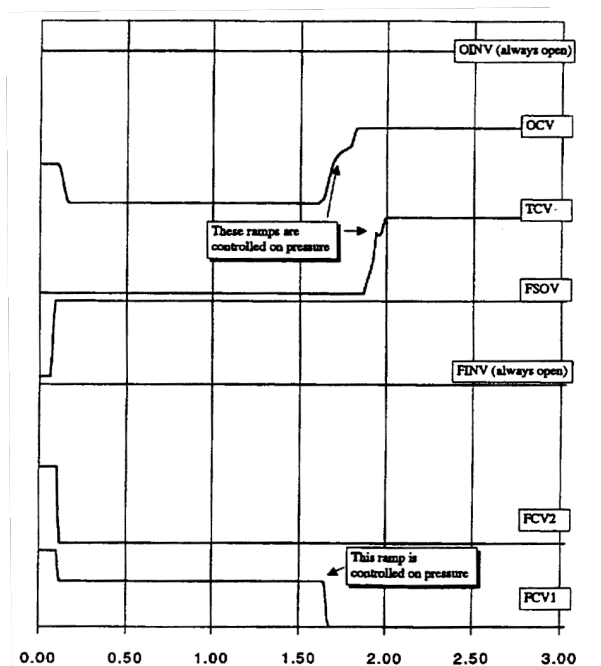


Figure 3. RL-10A-3-3A Valve schedule for Start-up Simulation³

Ignition of the main combustion chamber usually occurs approximately 0.3 seconds after the main-engine start signal ($t = 0$) is given (for first-burns). The ignition source is an electric spark, powering a torch igniter. The ignited combustion chamber provides more thermal energy to drive the turbine. As the turbopumps accelerate, engine pneumatic pressure is used to close the interstage cool-down valve completely and open the OCV at pre-set fuel and LOX pump discharge pressures. The OCV typically opens very quickly and the resultant flood of oxygen into the combustion chamber causes a sharp increase in system pressures. During this period of fast pressure rise, the thrust control valve (TCV) is opened, regulated by a pneumatic lead-lag circuit to control thrust overshoot. The engine then settles to its normal steady-state operating point.

III. Description of the transient model of the RL-10 engine

A. Modelling principles

The development of the RL-10 engine transient model was conducted with a commercial tool called Ecosim-Pro and the European Space Propulsion System Simulation library (ESPSS) which enables the modelling and analysis of propulsion systems for both spacecraft and launcher applications. The ESPSS is a set of libraries written to model all aspects of a propulsion system.

B. Fluid Properties

In a cryogenic rocket engine, the variations of fluid properties during the transient phase have a huge impact on the engine performances and behaviour, most notably on the coolant side: mass flow rate, pressure losses, heat transfers inside the cooling jacket, etc. The fluid properties library is in charge of the calculation of fluid properties. Functions available on this library are mainly used by the 1-D fluid flow library for the simulation of fluid systems. Real fluids properties are considered in the model for the couple LH_2 and LO_2 , by tables depending on both pressure and temperature derived from NIST database.¹¹ This class covers liquid, superheated, supercritical and two-phase fluids. Two-phase, two fluids mixtures of a real fluid in any thermodynamic state with a non-condensable gas (ideal) are allowed, as in the case of pre-start phase where propellants are in contact with gaseous helium. The homogeneous equilibrium model is used to

calculate the properties (quality, void fraction, etc..) of a real fluid in two-phase conditions, with or without a non-condensable gas mixture.

C. Turbomachinery modelling

1. Brief description

The fuel pump consists of two stages, separated by an interstage duct, which is vented via the interstage cool-down valve (FCV-1) during start. Both fuel pump stages have centrifugal impellers, vaneless diffusers and conical exit volutes; the first stage also has an inducer.

The LOX pump consists of an inducer and a single centrifugal impeller, followed by a vaneless diffuser and conical exit volute. The LOX pump is driven by the fuel turbine through the gear train. The turbopump speed sensor is located on the LOX pump shaft.

The RL-10 turbine is a two stage axial-flow, partial admission, impulse turbine. Downstream of the turbine blade rows, exit guide vanes reduce swirling of the discharged fluid. The turbine is driven by hydrogen and powers both fuel and oxidiser pumps.

There are a number of shaft seals which permit leakage from the pump discharge in order to cool the bearings. The fuel pump and the turbine are on a common shaft; power is transferred to the LOX pump through a series of gears. The seals, bearings, gear train all contribute to rotordynamic drag on the turbopump.

2. Modelling approach

Pumps

The pump model makes use of performance maps for head and resistive torque. The pump curves are introduced by means of fixed 1-D data tables defined as functions of a dimensionless variable θ that preserves homologous relationships in all zones of operation. θ parameter is defined as follows:

$$\theta = \pi + \arctan(\nu/n) \quad (1)$$

where ν and n are the reduced flow and speed parameters respectively:

$$\nu = \frac{Q}{Q_R} = \frac{\dot{m}_{in}/\rho_{in}}{Q_R} \quad n = \frac{30\omega/\pi}{rpm_R} \quad (2)$$

The dimensionless characteristics (head and torque) are defined as follows:

$$h = \frac{TDH/TDH_R}{n^2 + \nu^2} \quad \beta = \frac{\tau/\tau_R}{n^2 + \nu^2} \quad (3)$$

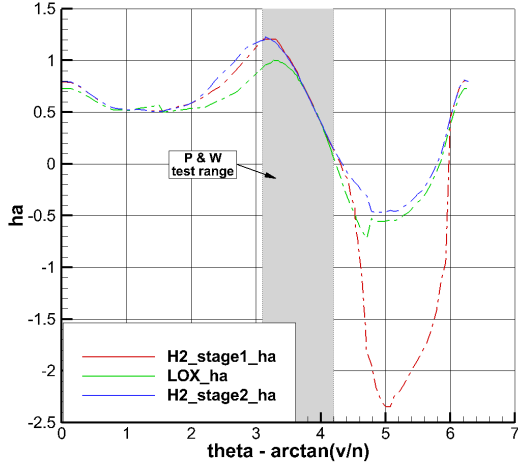
this method eliminates most concerns of zero quantities producing singularities. To simplify the comparison with generic map curves, these relations are normalized using the head, torque, speed and volumetric flow at the point of maximum pump efficiency. These maps are a combination of available test data provided by Pratt&Whitney³ and generic pump performance curves⁴ (see Figure 4, test data range in grey). Additional maps were established (not shown here), giving a corrective factor on the pump torque, function of the rotational speed ratio (also provided by P&W). The enthalpy flow rise is a function of the absorbed power while the evaluation of the mass flow rate is performed through an ODE.

Turbine

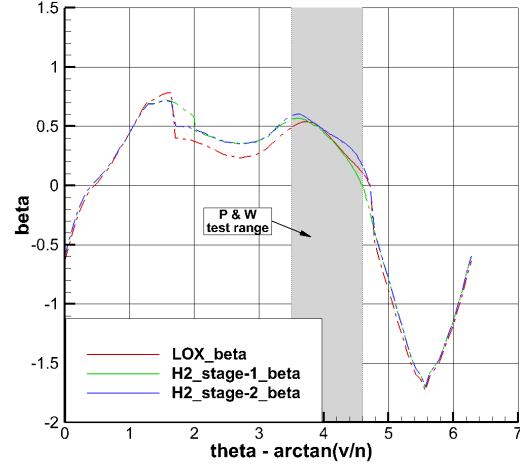
The turbine performance maps provided by Pratt&Whitney depict the combined performance of the two stages. The first one describes the effective area (area times discharge coefficient) as a function of velocity ratio (U/C_o) for several different pressure ratios. The second one describes the combined two-stage turbine efficiency as a function of velocity ratio (U/C_o) as well. In the present study, Pratt & Whitney performance maps are transformed to obtain the turbine performance maps used in the ESPSS turbine model. These maps (mass flow coefficient and specific torque) are introduced by means of 2-D input data tables as a function of velocity ratio and pressure ratio:

$$N = \frac{r \cdot \omega}{C_o} \quad \Pi = P_{01} / P_{02} \quad (4)$$

and the mass flow coefficient and specific torque are defined as:



(a) Extended Head map for LOX and Fuel pumps



(b) Extended Torque map for LOX and Fuel pumps

Figure 4. Pumps performance maps

$$Q^+ = \frac{\dot{m}_{map} \cdot C_o}{r^2 P_{01}} \quad ST = \frac{\tau}{r \dot{m}_{map} C_o} \quad (5)$$

D. Thrust chamber and cooling jacket modelling

The thrust chamber component, inherited from the original ESPSS library,⁷ represents a non adiabatic 1-D combustion process inside a chamber for liquid or gas propellants. The equilibrium combustion gases properties (molar fraction, thermodynamic and transport properties) are calculated for each chamber volume (node) using the minimum Gibbs energy method⁸ as a function of the propellants mixture molar fractions, inlet conditions and chamber pressure. Transient chamber conditions (pressures, temperatures, mass flows and heat exchanged with the walls) are derived from 1-D transient conservation equations. A mixture equation between the injected propellants and the combustion gases is applied. From the definition of the mixture ratio MR and derivation, the following dynamic equation gives the MR evolution:

$$\dot{m}_{ox} = MR \dot{m}_{fu} + \frac{d}{dt} (MR) \frac{\rho V_{ch}}{1 + MR} \quad (6)$$

Combustion take place when mixture ratio is within the allowed limits, the ignition flag is active and a minimum time (ignition delay) τ has elapsed. Mass, energy and momentum equations are basically the same as in the pipe component with variable cross area, Equations (7),(8).

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial f(\mathbf{u})}{\partial x} = S(\mathbf{u}) \quad (7)$$

where

$$\mathbf{u} = A \begin{bmatrix} \rho \\ \rho x^{nc} \\ \rho v \\ \rho E \end{bmatrix}; \quad f(\mathbf{u}) = A \begin{bmatrix} \rho v \\ \rho v x^{nc} \\ \rho v^2 + P \\ \rho v H \end{bmatrix}; \quad (8)$$

$$S(\mathbf{u}) = \begin{bmatrix} -\rho A k_{wall} (\partial P / \partial t) \\ -\rho x^{nc} A k_{wall} (\partial P / \partial t) \\ -0.5(d\xi/dx) \rho v |v| A + \rho g A + P(dA/dx) \\ \dot{q}_w(dA_{wet}/dx) + \rho g v A \end{bmatrix} \quad (9)$$

The centred scheme is used to discretise the chamber, using a staggered mesh approach(see Figure 5).

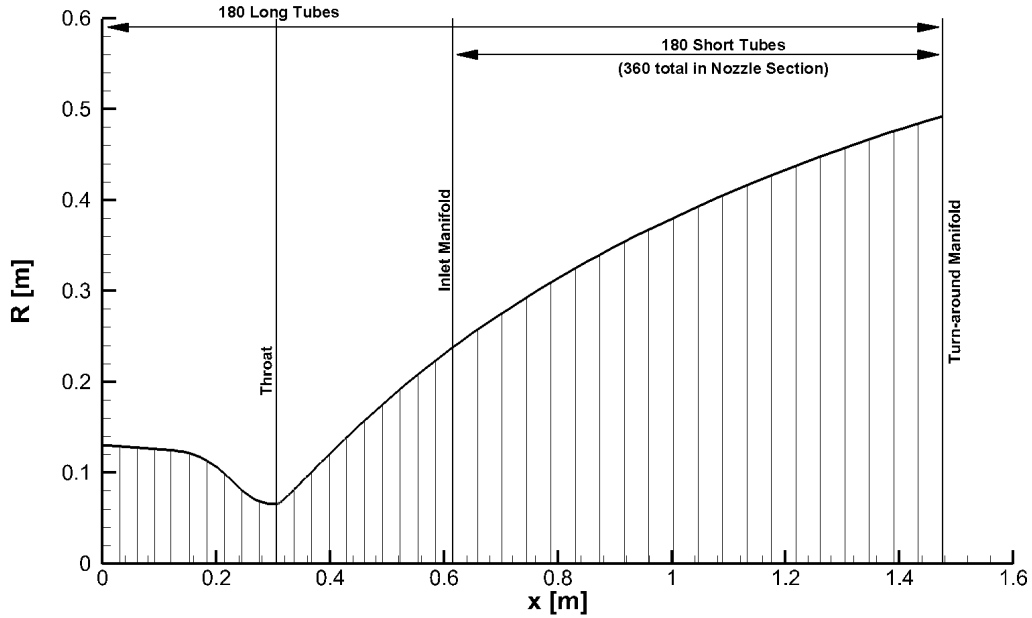


Figure 5. RL-10A-3-3A chamber contour³ and discretisation

The walls represented by thermal components in the Cooling Jacket component are not included in the chamber model, but are taken as a boundary for the heat exchange calculation instead:

$$\dot{q}_{wall} = h_c A_{wet} (T_{aw} - T_w) + \sigma A_{wet} (T_{core}^4 - T_w^4) \quad (10)$$

In the combustion chamber component the heat transfer coefficient h_c can be evaluated by different correlations (original Bartz equation, modified Bartz equation, Pavli equation). Please refer to the paper presented at Space Propulsion Conference 2010⁶ for a detailed description of the heat transfer correlation models. We chose the original Bartz equation to keep the computational time low. The Bartz equation has been rewritten in a Stanton type form:

$$St_{Bartz} = 0.026 \left(\frac{\mu_{ref}^{0.2}}{c_{p,ref}^{0.6}} \right) \left(\frac{\lambda_{ref}}{\mu_{ref}} \right)^{0.6} (\dot{m})^{-0.2} A^{0.1} \left(\frac{\pi/4}{R_{curv} D_{th}} \right)^{0.1} \quad (11)$$

The injector plate composed by injectors and injector domes is modelled by a component that takes into account the convective and radiative heat transfer between the fluid in the first volume of the chamber and the face plate, and evaluates the conductive and capacitive effect of the injector walls in an accurate way, representative of a generic injector head.⁶

The cooling jacket model is divided into a variable number of sections in axial direction. Every section is made of one fluid node of the “Tube” component (from FLUID_FLOW_1D library, see Equations (7),(8)), which is simulating the cooling channels and five slices of the “3D wall” components, which are simulating the metallic walls. The walls are divided in 5 different 3-D components as shown in Figure 6(a); the contours of the height and width of the RL-10 channels are shown in Figure 6(b). Each component has a 3-dimensional discretisation in tangential, radial and longitudinal direction (dx , dy , dz), respectively.

The combustion chamber model has been discretised in 40 volumes: 10 volumes for the subsonic part, 10 from the throat to the cooling jacket inlet manifold and 20 from the inlet manifold to the outlet (see Figure 5). The cooling jacket model is constructed of 360 stainless steel tubes of type 347SS. There are 180 short tubes, from inlet manifold to the turn-around one, and other 180 long tubes, from the turn-around manifold to the injector plate. The short and long tubes are arranged side-by-side in the nozzle section. Since the cooling channel shape is not rectangular but slightly rounded, a detailed geometrical reconstruction has been performed to assess the effective exposed surface area. To this purpose the Pratt&Whitney specification has been accepted regarding the angle of exposure which is around 112° .³

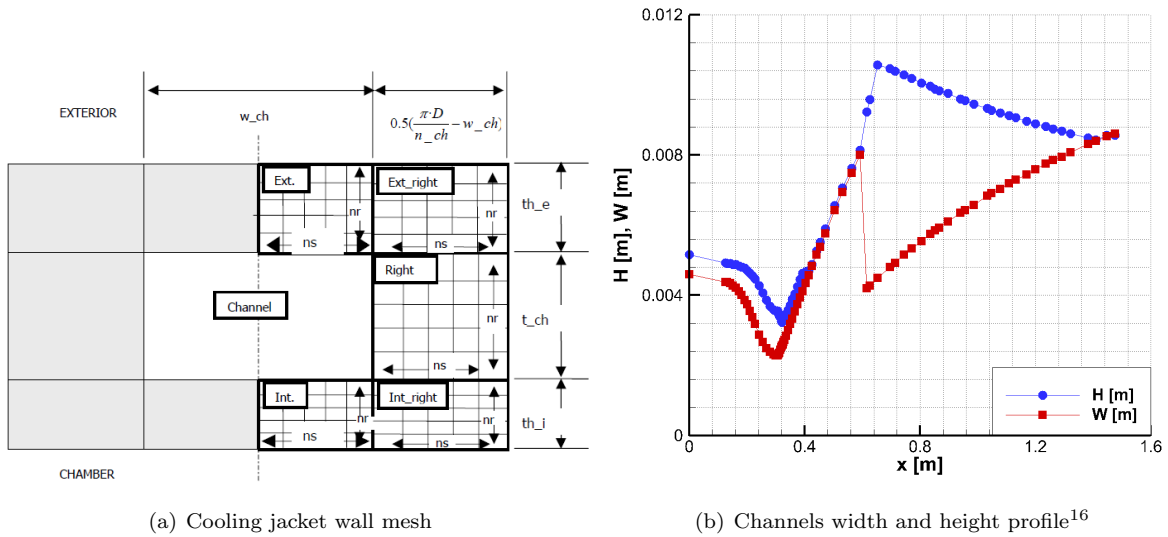


Figure 6. Cooling jacket channels and walls

E. Lines, valves and manifolds modelling

In addition to the various subsystem listed above, there are on the RL-10 engine a large number of lines valves and manifolds. Valves are modelled as zero dimensional components while the lines present in the engine are modelled via an area-varying non-uniform mesh 1-D scheme. Where possible and data were available a detailed geometrical reconstruction has been performed, as for the case of the Venturi pipe and the discharge turbine pipe. The governing equations are the same used to model the cooling channels (see Equations (7),(8)).

IV. Results

A. Subsystem design

Each component of the RL-10A-3-3A engine has been previously simulated as a stand-alone component to validate its behaviour at steady state conditions, then they have been grouped in several subsystems:

- Turbopump assembly
- Thrust chamber and cooling jacket
- Oxidiser pipe line

- Fuel pump to cooling jacket pipe line
- Cooling jacket to turbine pipe line
- Turbine to chamber pipe line

All the subsystem models have then been connected together to create the complete RL-10A-3-3A engine model. Two different configurations of the engine model have been adopted: the first one to match the engine nominal operation point and a second one to match the ground test results.

B. Steady state performance

Nominal operation point has been considered for the steady-state performance prediction. Flight data have not been considered in this comparison because insufficient data exist to determine the mixture ratio and trim position of the oxidiser control valve (OCV). Table 2 shows relative performance predictions of the transient model at steady state conditions. Where available, experimental values at the end of the transient phase have been used as reference;³ other performance parameters have been compared at their nominal operating condition.^{7,3,9}

Table 2. RL-10A-3-3A engine system output data

Name	Description	Error
P_{cc} [bar]	Chamber pressure	0.16%
MR [-]	Mixture Ratio	0.58%
$m_{cc,ox}$ [kg/s]	LOX chamber mass flow	0.38%
$m_{cc,fu}$ [kg/s]	H ₂ chamber mass flow	-0.77%
$m_{cc,t}$ [kg/s]	Total chamber mass flow	-0.09%
$T_{in,t}$ [K]	Turbine inlet temperature	4.31%
Π_{tt} [-]	Turbine pressure ratio	0.33%
ω_t [rpm]	Turbine rotational speed	0.015%
τ_t [N·m]	Turbine torque	1.3%
$\dot{m}_t$ [kg/s]	Turbine mass flow	1.35%
W_t [kW]	Turbine power	1.31%
$\dot{m}_{p,LOX}$ [kg/s]	LOX pump mass flow	0.38%
$\omega_{p,ox}$ [rpm]	LOX pump rotational speed	0.015%
$\tau_{p,ox}$ [N·m]	LOX pump torque	5.81%
$W_{p,ox}$ [W]	LOX pump power	5.8%
$\dot{m}_{p1,LH2}$ [kg/s]	H ₂ 1st stage mass flow	0.77%
$\dot{m}_{p2,LH2}$ [kg/s]	H ₂ 2nd stage mass flow	0.77%
$\omega_{p,LH2}$ [rpm]	H ₂ rotational speed	0.015%
$\tau_{p1,LH2}$ [N·m]	H ₂ 1st stage torque	2.17%
$\tau_{p2,LH2}$ [N·m]	H ₂ 2nd stage torque	0.16%
$W_{p1,LH2}$ [W]	H ₂ 1st stage power	2.18%
$W_{p2,LH2}$ [W]	H ₂ 2nd stage power	0.18%
$Thrust$ [kN]	Engine thrust	1.42%
Isp [s]	Engine specific impulse	0.69%

C. Start transient

The results of start transient simulations were compared with measured data of a single ground test first-burn (P2093 Run 3.02 - Test 463)³ and with the simulation results of a previous work.³ Since no detailed

initial conditions along the engine were available, a simulation of the pre-start phase was necessary to obtain reasonable initial conditions for the engine start.

The inlet pressure and temperature used in the model are coming from nominal operation conditions. Another important variable is the cooling jacket initial temperature, that has been set at 300 K before the pre-start phase occurs. After the pre-start simulation the cooling jacket wall temperature decreased around 240 K. It is clear that the cooling jacket wall temperatures have a great importance since they help to determine the engine start capability. The cooling jacket manifold has a lower temperature than the cooling jacket because it is partially filled by gaseous hydrogen that has not vented overboard via the fuel discharge valve (FCV-2).

In the simulation the ignition occurs when the propellant mixture ratio inside the combustion chamber reaches a value lower than 30 (as shown in Ref.¹⁴), at around 0.3 seconds as in the ground test.

Figure 8(a) shows the comparison between measured and predicted chamber pressure. The model matches the measured time-to-accelerate to within approximately 92 milliseconds (the “time-to-accelerate” is defined here as the time from 0 seconds at which the chamber pressure reaches 13.79 bar (200 psia)).

The presence of small oscillations evident in the test data are due to oscillations of the TCV servo-mechanism. Such a mechanism is absent in the model so no oscillations occur. To obtain a reasonable chamber pressure profile the TCV opening sequence has been modified (see Figure 7) using the opening sequence obtained from a dynamic model of the TCV valve as a guideline.¹³ With this new opening sequence the difference with the previous model of the RL-10A-3-3A transient start-up³ is remarkable.

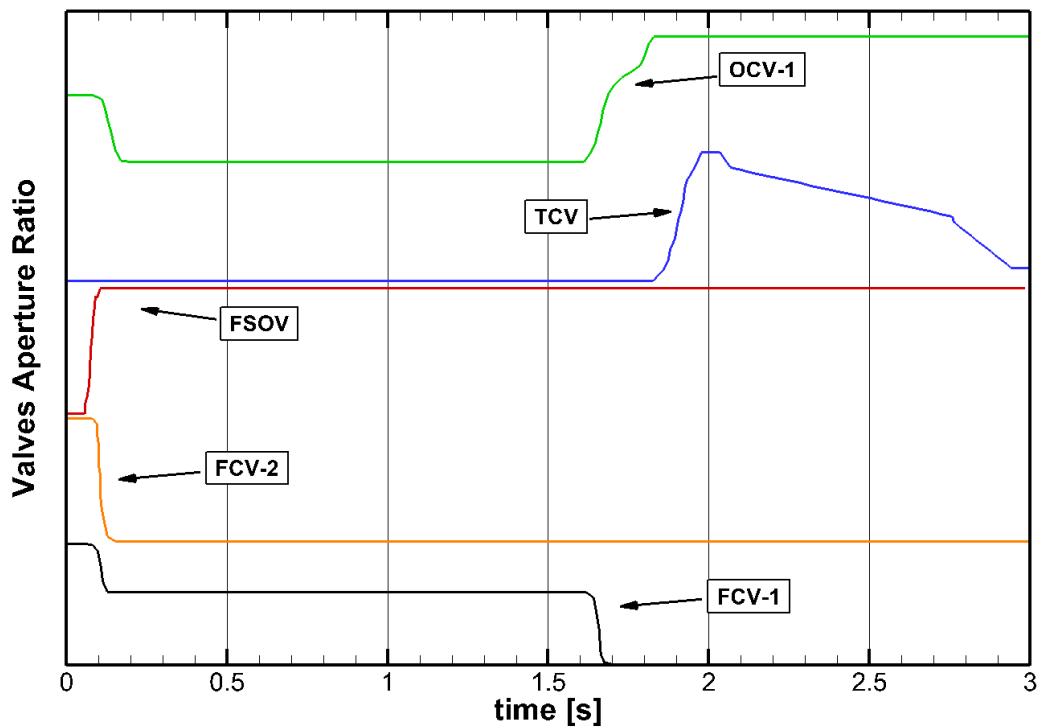
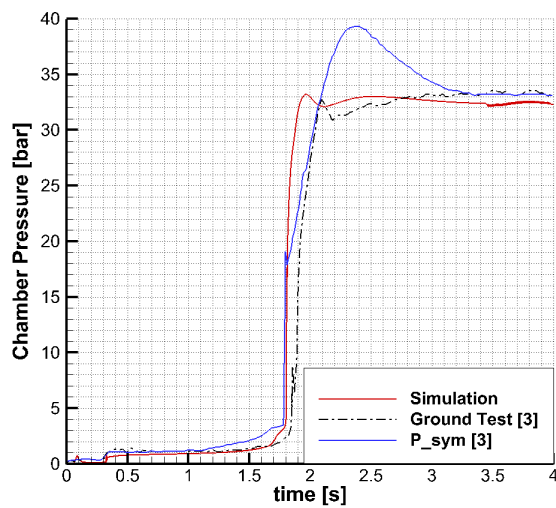


Figure 7. Valves opening sequence adopted in the simulation

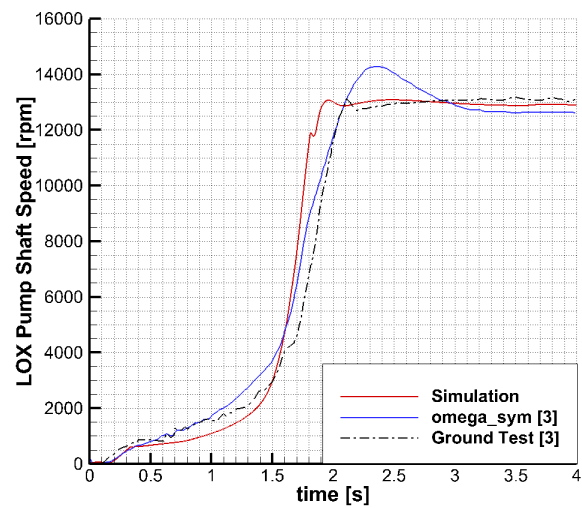
In figure 8(b) and figure 9(a,b) the simulation exhibits some sharp transient before reaching steady-state conditions; these seem to be due to fluid compressions and phase changes that occur when the OCV suddenly opens. These transients are steeper than the measured data probably because the dynamic behaviour of the OCV valve plays an important role in the fluid dynamics during pressure rise.

Figure 9(b) shows the fuel inlet mass flow trend; as for the chamber pressure, also for the measured hydrogen mass flow the evident oscillations are explained by the oscillations of the TCV servo mechanism.

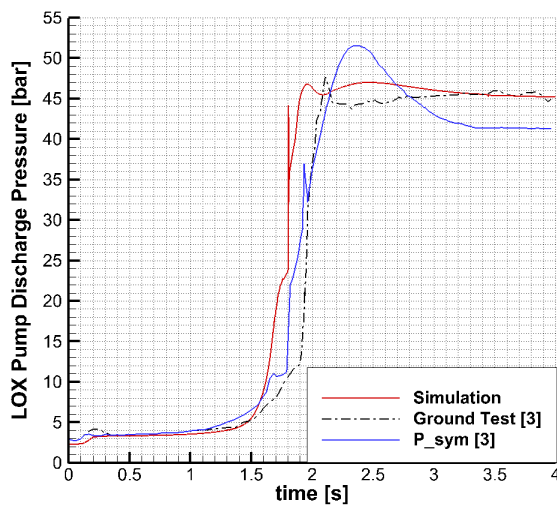
Unfortunately on the fuel side no turbopump measured data are available so no comparison has been possible between the simulated and experimental results.



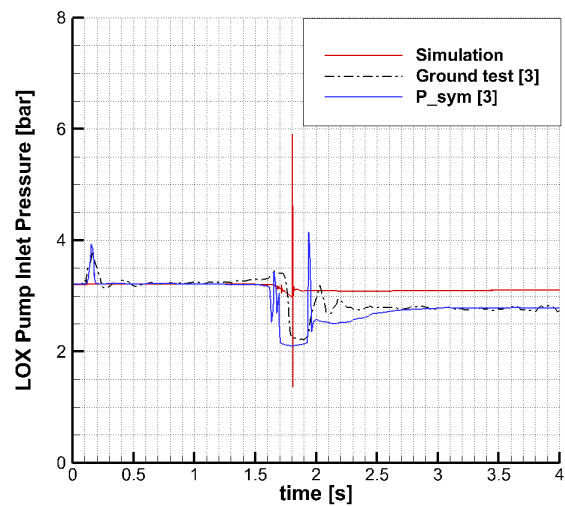
(a) Chamber Pressure



(b) LOX Pump Shaft Speed

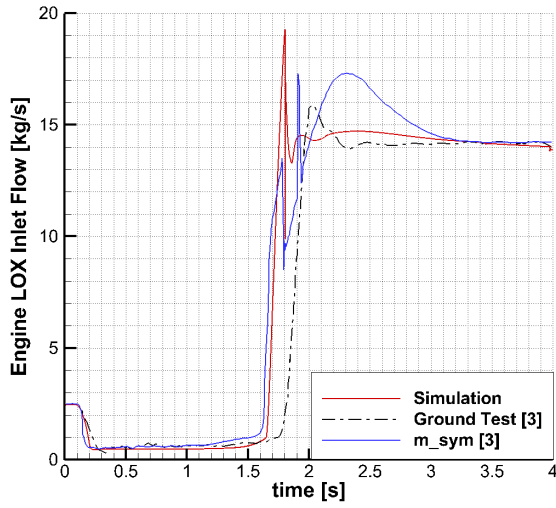


(c) LOX Pump Discharge Pressure

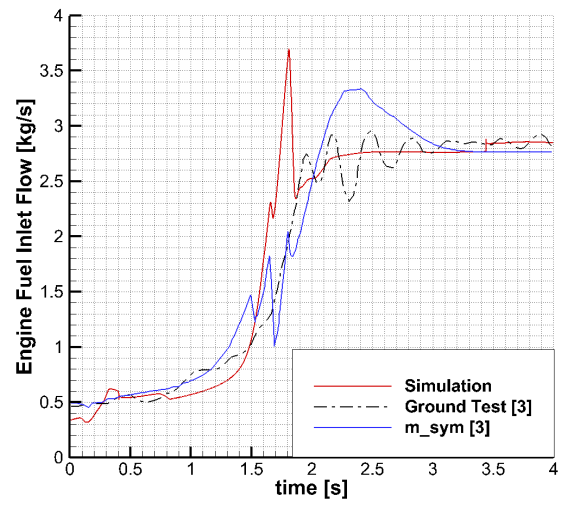


(d) LOX Pump Inlet Pressure

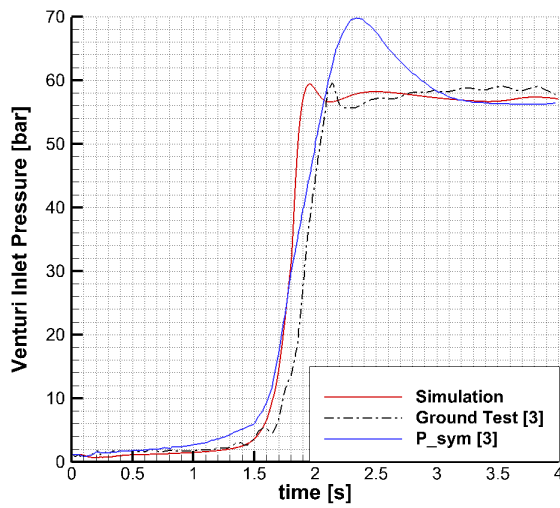
Figure 8. Transient results - part 1



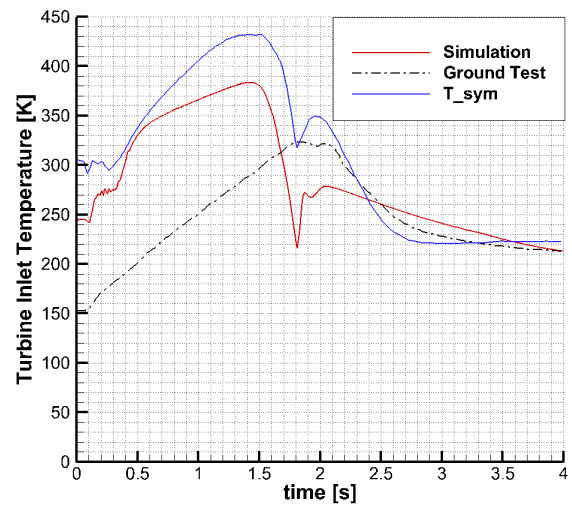
(a) LOX Engine inlet mass flow



(b) Fuel Engine inlet mass flow



(c) Venturi inlet Pressure



(d) Turbine Inlet Temperature

Figure 9. Transient results - part 2

V. Conclusion

The major goals set for this work were to create a transient model for liquid rocket engines, develop a procedure able to simulate and predict the start-up transient phase for future rocket engines, and to validate the model with an existing liquid rocket engine, the RL-10A-3-3A. These goals have been accomplished.

Comparison of the transient behaviour of the engine during ground test and model predictions is very satisfactory. Although many uncertainties affect the transient simulation (such as valve discharge coefficient uncertainties, running shaft torque, OCV behaviour, initial conditions uncertainties etc.) the model correctly reproduces the main phenomena occurring during transients, such as combustion, heat transfer, turbopump operation phase change, valve maneuvering and pressure drops, as well as the thermodynamic behaviour of the fluids. Two phase flow effects in the engine are also well estimated. Moreover the RL-10A-3-3A model accurately predicts the engine time-to-accelerate when compared to ground test data.

Improvements can still be done, especially in the direction of pressure-operated valves modelling, to obtain a better and more realistic control of the engine pressure and thrust build-up.

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